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**Project Report**

**Project Title: 4 Bit Magnitude Comparator**

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## Title:

### Design and Simulation of a 4-Bit Magnitude Comparator

## Introduction

A magnitude comparator is a combinational logic circuit used in digital electronics to compare two binary numbers and determine their relative magnitudes. It checks whether one number is greater than, less than or equal to another number. Comparators are widely used in microprocessors, arithmetic logic units (ALU), digital systems and decision-making circuits.

A 4-bit magnitude comparator compares two 4-bit binary numbers. Let the two numbers be:

- $A = A_3 A_2 A_1 A_0$
- $B = B_3 B_2 B_1 B_0$

The comparator produces three outputs:

- $A > B$  (A is greater than B)
- $A < B$  (A is less than B)
- $A = B$  (A is equal to B)

This project focuses on designing a 4-bit magnitude comparator using basic logic gates, simulating the circuit using software, and implementing the circuit in hardware.

## Objective

The main objectives of this project are:

1. To understand the working principle of a magnitude comparator.
2. To design a 4-bit magnitude comparator using logic gates and 74085 IC.
3. To create the truth table for comparison operations.
4. To simulate the circuit using simulation software.
5. To implement the circuit using hardware components.
6. To analyze the output results of the comparator.

## Project Description

A 4-bit magnitude comparator compares two 4-bit binary numbers bit by bit starting from the most significant bit (MSB).

Comparison logic:

1. If  $A_3 > B_3 \rightarrow A > B$
2. If  $A_3 = B_3$ , compare  $A_2$  and  $B_2$
3. Continue until a difference is found
4. If all bits are equal  $\rightarrow A = B$

Outputs:

- $G$  (Greater)  $\rightarrow A > B$

- E (Equal)  $\rightarrow A = B$
- L (Less)  $\rightarrow A < B$

#### i. Block Diagram

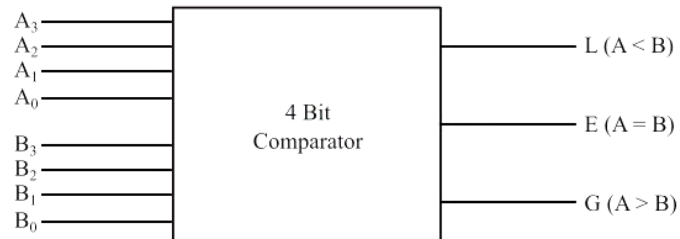


Figure: Block Diagram

#### ii. Truth Table

For, 1-bit Comparator  $\rightarrow$  2 variables  $\rightarrow$  4 rows

And, 2-bit Comparator  $\rightarrow$  4 variables  $\rightarrow$  16 rows

Therefore, n-bit Comparator  $\rightarrow$  2n variables  $\rightarrow 2^{2n}$  rows

so, 4-bit Comparator  $\rightarrow 2 \times 4 = 8$  variables  $\rightarrow 2^{2 \times 4} = 256$  rows

A full 4-bit comparator has 256 combinations ( $2^8$ ), which is too large to display entirely. So it is usually represented using pattern-based truth logic:

#### Truth Table (for 1-bit comparator)

| A | B | A = B | A > B | A < B |
|---|---|-------|-------|-------|
| 0 | 0 | 1     | 0     | 0     |
| 0 | 1 | 0     | 0     | 1     |
| 1 | 0 | 0     | 1     | 0     |
| 1 | 1 | 1     | 0     | 0     |

#### Expanded Pattern Table (MSB Priority)

| A3 | B3 | A2 | B2 | A1 | B1 | A0 | B0 | Output |
|----|----|----|----|----|----|----|----|--------|
| 1  | 0  | X  | X  | X  | X  | X  | X  | A > B  |
| 0  | 1  | X  | X  | X  | X  | X  | X  | A < B  |
| 0  | 0  | 1  | 0  | X  | X  | X  | X  | A > B  |

|            |            |            |            |            |            |            |            |       |
|------------|------------|------------|------------|------------|------------|------------|------------|-------|
| 0          | 0          | 0          | 1          | X          | X          | X          | X          | A < B |
| 0          | 0          | 0          | 0          | 1          | 0          | X          | X          | A > B |
| 0          | 0          | 0          | 0          | 0          | 1          | X          | X          | A < B |
| 0          | 0          | 0          | 0          | 0          | 0          | 1          | 0          | A > B |
| 0          | 0          | 0          | 0          | 0          | 0          | 0          | 1          | A < B |
| Equal bits | Equal bits | Equal bits | Equal bits | Equal bits | Equal bits | Equal bits | Equal bits | A = B |

**Note:**

- X = don't care
- Comparison starts from  $A_3$  vs  $B_3$  (MSB)

### iii. Boolean Equations

- **(A = B) Equality Output**

For 1-bit comparator, Output equation =  $AB + A'B' = A \odot B$

Similarly for 4-bit comparator,

A is equal to B only when all bits are equal:

$$A_3 A_2 A_1 A_0 \quad B_3 B_2 B_1 B_0$$

$$E = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$$

Where:  $\odot$  = XNOR operation

- **(A > B) Greater Output**

For 1-bit comparator, Output equation =  $AB'$

Similarly for 4-bit comparator,

$$A_3 > B_3$$

$$\text{OR } A_3 = B_3 \text{ AND } A_2 > B_2$$

$$\text{OR } A_3 = B_3 \text{ AND } A_2 = B_2 \text{ AND } A_1 > B_1$$

$$\text{OR } A_3 = B_3 \text{ AND } A_2 = B_2 \text{ AND } A_1 = B_1 \text{ AND } A_0 > B_0$$

$$G = A_3 B_3' + (A_3 \odot B_3) A_2 B_2' + (A_3 \odot B_3)(A_2 \odot B_2) A_1 B_1' + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1) A_0 B_0'$$

- **(A < B) Less Output**

For 1-bit comparator, Output equation =  $A'B$

Similarly for 4-bit comparator,

$$A_3 < B_3$$

$$\text{OR } A_3 = B_3 \text{ AND } A_2 < B_2$$

$$\text{OR } A_3 = B_3 \text{ AND } A_2 = B_2 \text{ AND } A_1 < B_1$$

$$\text{OR } A_3 = B_3 \text{ AND } A_2 = B_2 \text{ AND } A_1 = B_1 \text{ AND } A_0 < B_0$$

$$L = A'_3B_3 + (A_3 \odot B_3)A'_2B_2 + (A_3 \odot B_3)(A_2 \odot B_2)A'_1B_1 + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)A'_0B_0$$

#### iv. Hardware Requirements

- IC 7485 (4-bit magnitude comparator)
- Breadboard (830 Pin)
- Jumper wires (Male-Male)
- LED (3 pieces for outputs)
- Resistors (220 $\Omega$ )
- Power supply (5V battery)
- Battery Case (1 piece)
- 3 Pin Switches (8 Pieces)

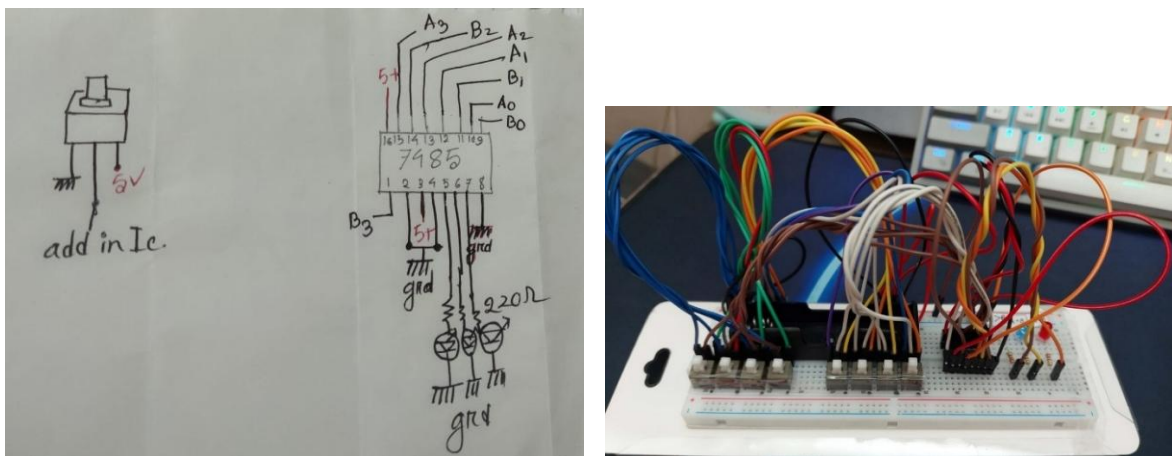


Figure: Hardware Implementation

#### v. Software Requirements

- CircuitVerse – Used for designing logic gate circuit of the 4-bit magnitude comparator.

- Proteus – Used for circuit simulation.
- Logic Gates – Required for circuit design:
  - NOT Gate
  - XNOR Gate
  - AND Gate
  - OR Gate
- Wire Connections – For connecting different components in the simulation environment.
- Inputs and Outputs Configuration:
  - 8 Inputs ( $A_3, A_2, A_1, A_0, B_3, B_2, B_1, B_0$ )
  - 3 Outputs ( $A > B, A = B, A < B$ )
- IC 7485 – Used in simulation tools for comparator implementation.

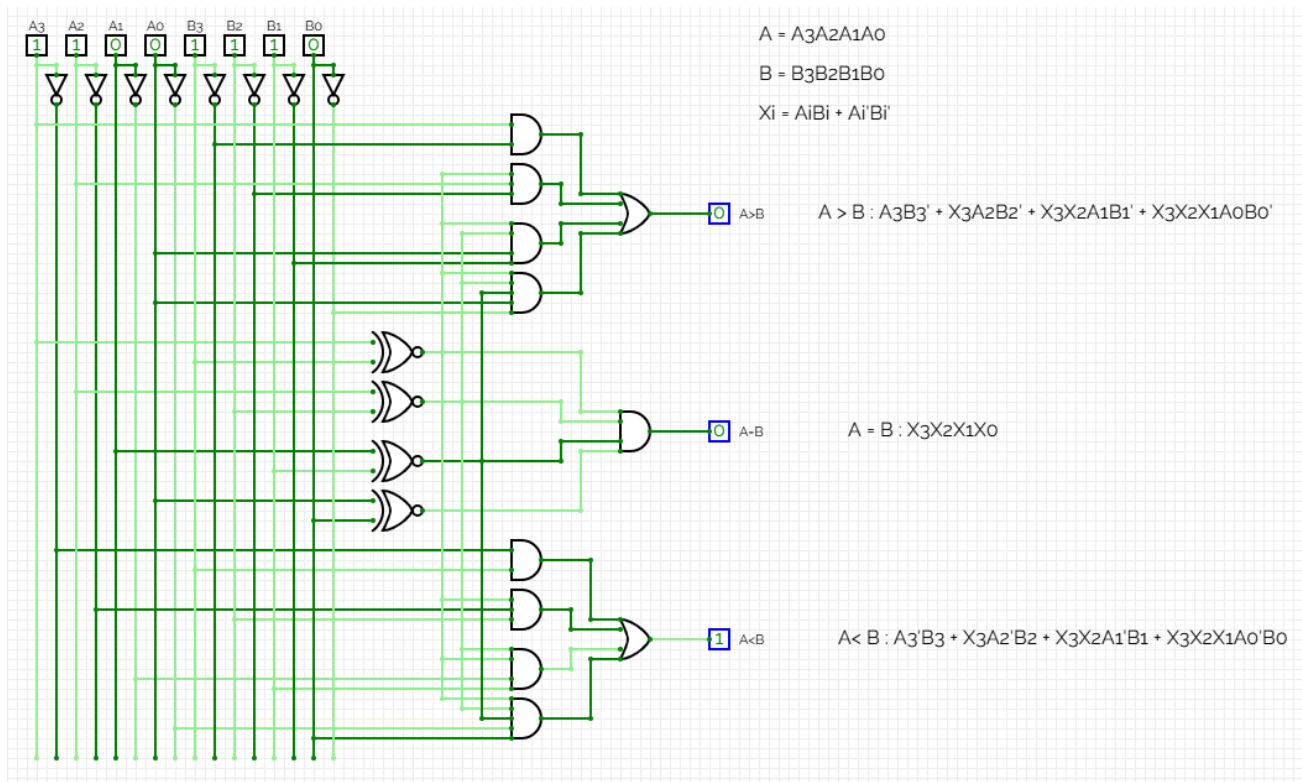


Figure: Logic Circuit Design

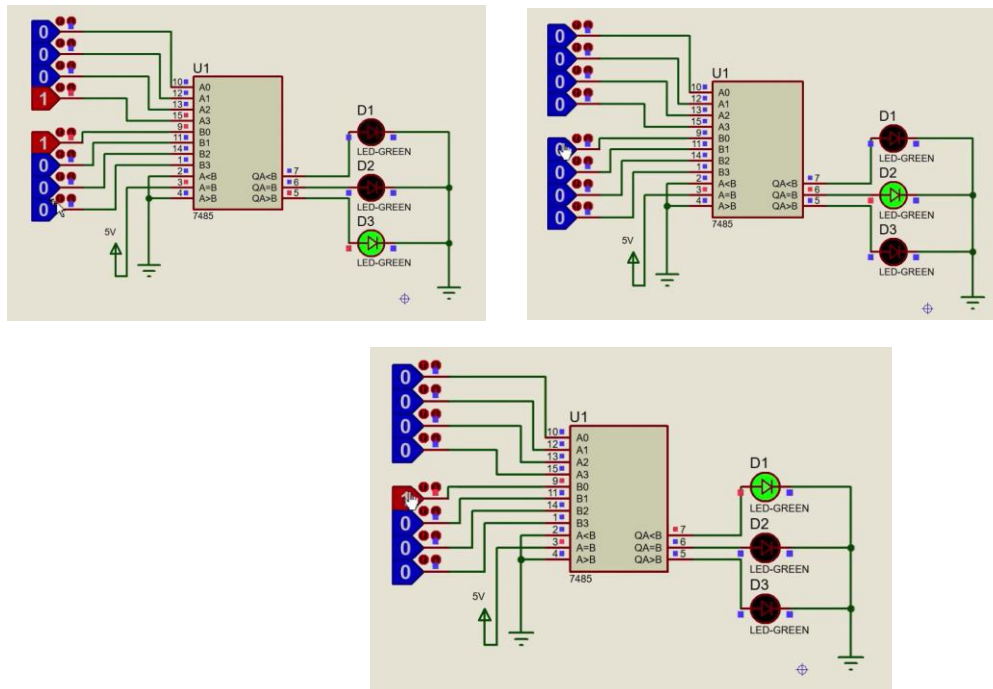


Figure: Software Simulation using 7485 IC

## Result Analysis

The simulation and hardware implementation successfully demonstrated the functionality of the 4-bit magnitude comparator. The LEDs correctly indicated the comparison result based on the input values. When the value of A was greater than B, the A > B LED turned ON. Similarly, when A was less than B, the A < B LED turned ON. When both values were equal, the A = B LED turned ON.

The circuit responded quickly to input changes, confirming that the comparator operates as a combinational logic circuit without memory.

## Discussion

The 4-bit magnitude comparator is an essential digital circuit used in many computing and control systems. Through this project, the design and working of comparator circuits were clearly understood. Both simulation and hardware implementation verified the theoretical concepts of digital comparison.

Future improvements may include:

- Expanding the comparator to 8-bit or 16-bit systems
- Using programmable logic devices (PLD)
- Integrating the comparator into arithmetic logic units (ALU)

## Conclusion

The project successfully designed, simulated and implemented a 4-bit magnitude comparator. The outputs correctly identified whether one binary number was greater than, less than or equal to another. Therefore, this experiment enhanced understanding of digital logic design and comparator circuits which are fundamental components in modern digital electronics.